

3° cell 2021

Ex 1

$$V_{CC} - I_C(1 + \frac{1}{\beta_F})R_E + V_{BE} - \frac{I_C R_B}{\beta_F} = 0$$

$$I_C = \frac{V_{CC} + V_{BE}}{\frac{R_E(1 + \frac{1}{\beta_F}) + R_B}{\beta_F}} = 0.744 \text{ mA}$$

$$V_{GS} = I_C R_C - I_D R_S \rightarrow I_D = \frac{I_C R_C - V_{GS}}{R_S} = I_{DSS} \left(\frac{1 - V_{GS}}{V_t} \right)^2 \rightarrow \begin{cases} V_{GS} = -22 \text{ V} \\ V_{GS} = 6.45 \text{ V} \end{cases} \text{ ok}$$

$$I_D = 0.99 \text{ mA}$$

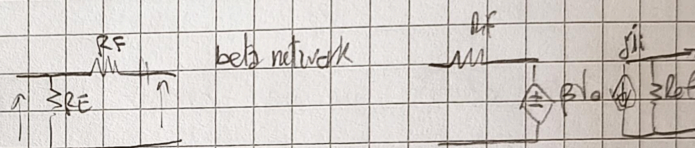
$$V_{DS} = V_{CC} - I_D(R_D + R_S) = 7.278 \text{ V}$$

$$V_{CE} = -(V_{CC} - R_E I_C(1 + \frac{1}{\beta_F}) - I_C R_C) = I_C R_C + R_E I_C(1 + \frac{1}{\beta_F}) - V_{CC} = -6.658 \text{ V}$$

$$Q_1(I_C, V_{CE}) = (0.744 \text{ mA}, -6.658 \text{ V}) \quad \checkmark$$

$$Q_2(I_D, V_{DS}) = (0.99 \text{ mA}, 7.278 \text{ V}) \quad \checkmark$$

Small signals: feedback topology (shooting caps)
voltage mixing (series) voltage sampling (shunt)
voltage divider
Thévenin input, single loop



$$R_{if} = R_E || R_F = 726 \Omega$$

$$R_{of} = R_E + R_F = 3.4 \text{ k}\Omega$$

$$\beta = \frac{R_{if}}{R_{of}} = \frac{R_E}{R_E + R_F} = 0.353$$

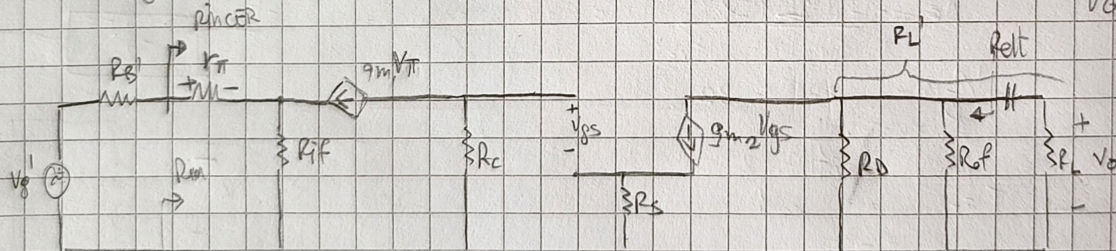
$$g_{m1} = \frac{I_C}{V_t} = 0.0298 \text{ S}^{-1}$$

$$r_{\pi} = \frac{\beta_0}{g_{m1}} = 3.356 \text{ k}\Omega$$

$$g_{m2} = \frac{2I_D}{V_{GS} - V_t} = 6.65 \cdot 10^{-4} \text{ S}^{-1}$$

$$V_g' = V_g \frac{R_B}{R_g + R_B} = V_g \cdot 0.9945$$

$$R_B' = R_B || R_g = 9.945 \text{ k}\Omega$$



$$A_a = \frac{V_o}{V_g'} \Big|_{\beta=0}$$

cont'd

$$R_{inCER} = r_{\pi} + (1 + \beta_0) R_{if} = 81.73 \text{ k}\Omega$$

$$R_{L'} = R_D || R_F || R_L = 2.216 \text{ k}\Omega$$

$$\alpha_{CS} = \frac{-g_{m2} R_{L'}}{1 + g_{m2} R_S} = -0.682 \text{ [V/V]}$$

$$\alpha_{CER} = \frac{-\beta_0 R_C}{R_{inCER}} = -12.235 \text{ [V/V]}$$

$$\alpha_i = \frac{R_{inCER}}{R_B' + R_{inCER}} = 0.887 \text{ [V/V]}$$

$$A_a = \alpha_i \alpha_{CER} \alpha_{CS} = +7.48 \text{ [V/V]}$$

$$A_F = \frac{V_o}{V_g'} = \frac{A_a}{1 + \beta A_a} = 2.06 \text{ [V/V]}$$

$$A_V = \frac{V_o}{V_g} = \frac{A_F R_B}{R_g + R_B} = 2.05 \text{ [V/V]}$$

$$R_{out}^a = R_{L'}$$

$$R_{out}^F = \frac{R_{out}^a}{1 + \beta A_a} = 813 \Omega$$

$$R_{out} = \frac{R_1}{1 - \frac{1}{R_1}} = 516 \Omega$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_g R_{gsc}} + \frac{1}{C_L R_L^{sc}} + \frac{1}{C_F R_F^{sc}} \right)$$

$C_g \ll C_L = C_F$
dominant part

$$= \frac{1}{2\pi} \frac{1}{C_g R_{gsc}}$$

no feedback
 $C_g \ll C_L = C_F$

$$R_{in}^{out} = R_B' + R_{in}^{out} = 91.7 k\Omega$$

$$R_{in}^F = R_{in}^2 (1 + \beta A_{vL}) = 331 k\Omega$$

$$R_{in} = R_{in}^F + R_B' = 321 k\Omega$$

$$R_{gsc} = R_g + R_B \parallel R_{in} = 282.6 k\Omega$$

$$f_L = \frac{1}{2\pi} \frac{1}{C_g R_{gsc}} = 58.3 Hz$$

considering less relevant components: $f_L = 58.3 + 28$

$$R_F^{sc} = R_D \parallel R_L + R_F + R_E \parallel (R_{in} + R_B') = 8.685 k\Omega$$

$$R_L^{sc} = R_{out}^F + R_L = 102 k\Omega$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_g R_{gsc}} + \frac{1}{C_L R_L^{sc}} + \frac{1}{C_F R_F^{sc}} \right) = 72 Hz$$

$$R_L^{sc} = R_D \parallel R_{out}^F + R_L = 102 k\Omega$$

$$R_F^{sc} = R_F + R_E \parallel (R_{in} + R_B') = 2.318 k\Omega$$

C_F

$$R_F^{sc} = R_F + R_E + R_D \parallel R_L = 9.77 k\Omega$$

$$R_{gsc} = R_g + R_B \parallel R_{in}^{out} = 88.18 k\Omega$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_g R_{gsc}} + \frac{1}{C_L R_L^{sc}} + \frac{1}{C_F R_F^{sc}} \right) = 267 Hz$$

$$f_L^{CL} = 76 Hz$$

Ex 2

$$R_1 = 18 k\Omega$$

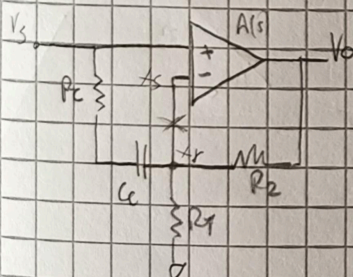
$$R_2 = 56 k\Omega$$

$$A(s) = \frac{A_0}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

$$A_0 = 10^5$$

$$\omega_{p1} = 10^2 rad/s$$

$$\omega_{p2} = 5 \cdot 10^5 rad/s$$



$$\beta = R_1 / (R_1 + R_2)$$

$$T = \frac{-x_r}{x_s} = \frac{A(s) \cdot R_1}{R_1 + R_2}$$

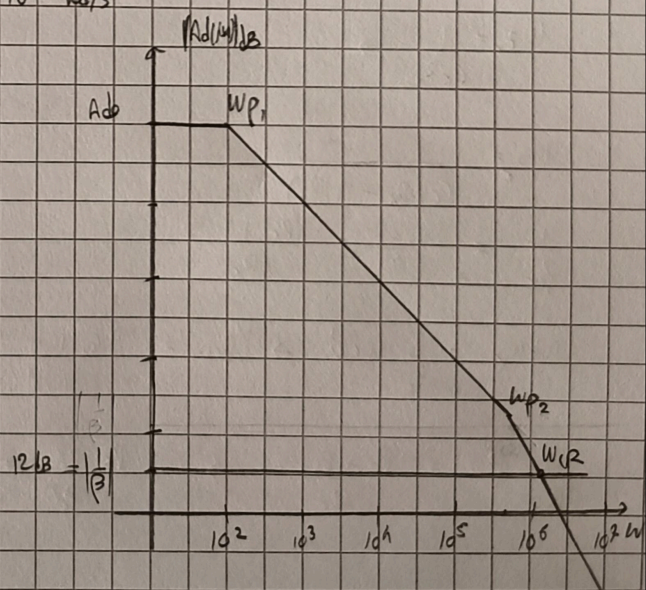
$$\beta = \frac{R_1}{R_1 + R_2} = 0.243$$

$$\frac{1}{\beta} = 4.1 = 12.2 dB$$

$$\begin{cases} A_0 \omega_{p1} = \omega_{p2} A\beta \\ A\beta \omega_{p2}^2 = \frac{1}{\beta} \omega_{cr}^2 \end{cases}$$

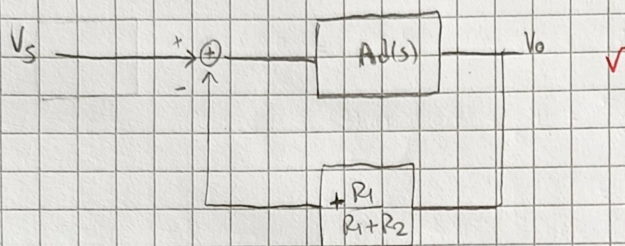
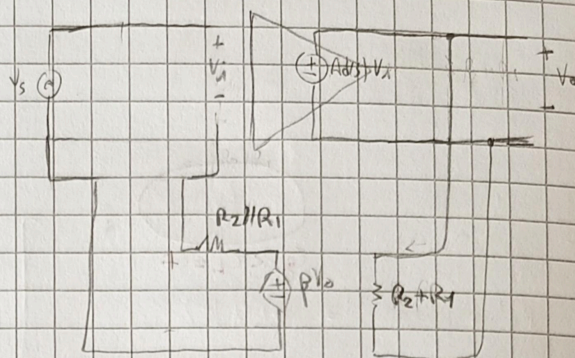
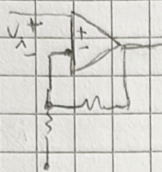
$$\omega_{cr} = \sqrt{A_0 \omega_{p1} \omega_{p2} \beta} = 1.102 \cdot 10^6 rad/s$$

$$\phi_M = 180 - 2 \arctan \left(\frac{\omega_{cr}}{\omega_{p1}} \right) - 2 \arctan \left(\frac{\omega_{cr}}{\omega_{p2}} \right) = 24.4^\circ \leftarrow \text{low phase margin}$$



$R_c = \infty$. Block diagram feedback theory $\rightarrow$ non inverting configuration, voltage mixing, voltage sampling $\rightarrow$ voltage divider and feedback

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$



expression of closed loop gain $\rightarrow AF = \frac{A_d}{1 + \beta A_d} = \frac{A_d(s)}{1 + \beta_c A_d} \xrightarrow{T \gg 1} \frac{1}{\beta_c}$

$$\beta_c = \beta_o \cdot \frac{1}{1 + sC \cdot R_c} = \frac{R_1}{R_1 + R_2} \cdot \frac{(1 + sC R_c)}{1 + sC(R_c + R_1 \parallel R_2)} \rightarrow \frac{1}{\beta_c} = \left(\frac{R_1 + R_2}{R_1} \right) \cdot \left[\frac{1 + sC(R_c + R_1 \parallel R_2)}{(1 + sC R_c)} \right]$$

$$R_c^{sc} = R_c + R_1 \parallel R_2$$

R_c to get 45° phase margin

$$W_{CR} = \frac{W_{p2}}{10} = 5 \cdot 10^4 \text{ rad/s}$$

$$W_{Z_c} = W_{CR}$$

$$W_{Z_c} = \frac{1}{C_c R_c}$$

$$W_{Z_c} = \frac{1}{C_c (R_c + R_1 \parallel R_2)}$$

$$\frac{1}{\beta_c \omega} = \left(\frac{R_1 + R_2}{R_1} \right) \cdot \frac{(R_c + R_1 \parallel R_2)}{(R_c)}$$

$$A_{d0} W_{p1} = \frac{1}{\beta_c \omega} W_{CR} \rightarrow \sqrt{\beta_c \omega} = A_{d0} \frac{W_{p1}}{W_{CR}} = 100$$

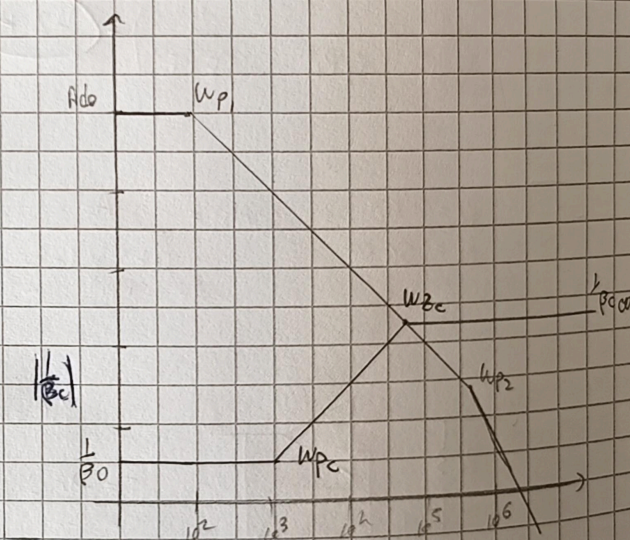
$$\frac{1}{\beta_c \omega} = \frac{R_c + R_1 \parallel R_2}{R_c} \rightarrow \frac{R_c}{\beta_c \omega} = R_c + R_1 \parallel R_2 \rightarrow R_c = \frac{R_1 \parallel R_2}{\frac{\beta_o - 1}{\beta_o}} = 286 \Omega$$

$$C_c = \frac{1}{R_c W_{Z_c}} = \frac{1}{R_c W_{CR}} = 70 \text{ nF}$$

$$\rightarrow W_{p_c} = 1.03 \cdot 10^3 \text{ rad/s}$$

$$PM = 180 - \arctan\left(\frac{W_{p1}}{W_{p2}}\right) - \arctan\left(\frac{W_{CR}}{W_{p2}}\right) - \arctan\left(\frac{W_{p1}}{W_{p_c}}\right) + \arctan\left(\frac{W_{p1}}{W_{Z_c}}\right)$$

$$\checkmark \quad \boxed{40.6^\circ}$$



Ado
e 20 min